

Human Trust Calibration in Human–Agent Collaboration for Autonomous Research

Abstract

The emergence of large language models (LLMs), tool-using agents, and multi-agent systems is transforming artificial intelligence from a passive information-processing technology into an active collaborator capable of planning, searching, coding, experimenting, analyzing results, and proposing scientific hypotheses. In autonomous research, however, technical capability alone does not determine whether collaboration will be effective. A central human-factors problem is **trust calibration**: ensuring that a researcher relies on an AI agent when its recommendations are dependable, questions or overrides it when it is unreliable, and maintains an accurate mental model of its capabilities and limitations. Traditional automation research distinguishes appropriate reliance from both over-trust and under-trust, while recent human-AI research emphasizes the importance of mental models, transparency, uncertainty communication, and adaptive oversight [1–5]. This paper examines how these ideas apply to autonomous research agents, where errors can propagate across literature retrieval, hypothesis generation, experimental design, code execution, interpretation, and scientific communication. Recent systems such as The AI Scientist and AI co-scientist demonstrate increasing levels of research autonomy, but also expose new trust-calibration requirements [6,7]. The paper synthesizes literature on automation trust, human-AI teaming, explainability, uncertainty, appropriate reliance, and agentic scientific discovery. It proposes a conceptual framework in which trust calibration is treated as a dynamic, evidence-based process linking agent capability, uncertainty, task criticality, transparency, human expertise, interaction history, and intervention policies. Key research gaps include the absence of standardized trust-calibration benchmarks for scientific agents, insufficient longitudinal studies, weak measurement of appropriate reliance, limited understanding of multi-agent trust, and inadequate mechanisms for communicating compound uncertainty. Future research should move beyond static confidence displays toward adaptive oversight, capability-aware delegation, provenance-centered interfaces, calibrated uncertainty communication, and human-agent shared mental models.

Keywords: trust calibration; human-agent collaboration; human-AI teaming; autonomous research; AI agents; scientific discovery; appropriate reliance; human oversight; large language models; trustworthy AI

1. Introduction

Artificial intelligence is increasingly being incorporated into scientific research as more than a computational support tool. Conventional scientific software generally performs well-defined functions such as statistical analysis, simulation, optimization, or information retrieval. In contrast, contemporary AI agents can combine language reasoning, external tools, memory, planning, code execution, web search, and iterative feedback to perform sequences of research activities with comparatively limited human intervention [8,9]. This

transition has been described as a movement from AI for science toward increasingly agentic scientific discovery [10].

The scientific research process is particularly attractive for agentic AI because it contains many activities that can be decomposed into interconnected subtasks. An autonomous research agent can potentially retrieve and synthesize literature, identify gaps, formulate hypotheses, generate experimental protocols, write and execute code, analyze results, and iteratively refine its research strategy. Recent systems illustrate this trajectory. The AI Scientist demonstrated an automated pipeline for generating research ideas, implementing experiments, analyzing results, and producing scientific papers in machine learning [6]. Similarly, Google's AI co-scientist was designed as a multi-agent system that generates, debates, ranks, and evolves scientific hypotheses in collaboration with human researchers [7].

Greater autonomy, however, changes the human-factors problem. When AI produces a recommendation, the researcher must decide whether to accept it. When an agent performs a long sequence of actions autonomously, the researcher must also decide **how much authority to delegate, when to inspect intermediate results, and when intervention is necessary**. These decisions are fundamentally related to trust.

Trust in automation has traditionally been understood as an attitude or expectation that an automated system will assist an individual in achieving a goal under conditions of uncertainty and vulnerability [1,11]. Importantly, trust is not synonymous with system reliability or with the user's willingness to use a system. The critical objective is **appropriate reliance**: reliance should increase when automation is capable and decrease when it is likely to fail [1,2]. Excessive trust can produce automation misuse or over-reliance, whereas insufficient trust can lead to disuse and lost benefits [2].

Autonomous research creates an especially difficult environment for trust calibration. Scientific tasks involve open-ended goals, incomplete information, uncertain evidence, heterogeneous sources, and consequences that may emerge only after several reasoning or execution steps. An agent may produce a plausible but incorrect literature summary, cite a nonexistent paper, write code containing a subtle methodological error, or formulate a scientifically interesting hypothesis based on a misleading premise. LLM hallucination and factuality problems therefore pose a direct challenge to calibrated reliance [12–14]. A 2024 empirical study of LLM-generated references for systematic reviews, for example, found substantial hallucination and low retrieval precision, demonstrating why generated scholarly claims and references require independent verification [13].

The central argument of this paper is that **trust calibration for autonomous research should not be treated merely as making users trust AI more or less**. Instead, it should be treated as a continuous human-agent coordination problem in which the human develops an evidence-based mental model of the agent's capabilities, limitations, uncertainty, and operating boundaries. The objective is to establish a collaboration regime in which the agent receives autonomy proportional to demonstrated competence and task risk, while humans retain meaningful authority over consequential scientific decisions.

This paper therefore reviews the foundations of trust calibration, connects them to human-AI and human-autonomous teaming, examines their relevance to autonomous scientific research, analyzes current approaches, and identifies research gaps and future directions.

2. Background and Related Work

2.1 Trust in Automation

The theoretical foundations of trust calibration originate largely in human factors and automation research. Muir conceptualized trust in automation as an important determinant of human intervention in automated systems, emphasizing the role of predictability, dependability, and faith in automation [11]. Parasuraman and Riley subsequently distinguished among **use, misuse, disuse, and abuse** of automation [2]. Misuse involves excessive reliance on automation, whereas disuse involves inappropriate rejection or non-use. These concepts remain highly relevant to AI agents.

Lee and See provided one of the most influential frameworks for trust in automation by arguing that the design objective should be **appropriate reliance**, rather than maximizing trust [1]. Their framework distinguishes between the trust that a person has in an automation and the actual capability of that automation. Calibration occurs when the human's trust appropriately corresponds to the system's capabilities.

A simplified relationship can be expressed as:

$$\text{Trust Calibration Error} = |\text{Human Trust} - \text{Perceived/Actual Capability}|$$

This formulation is conceptual rather than a universal measurement equation. It nevertheless captures the central idea: if a human believes an unreliable system is highly capable, over-trust results; if a highly capable system is systematically rejected, under-trust results.

Later research has identified multiple influences on trust, including system performance, transparency, reliability information, workload, task characteristics, user experience, and environmental conditions [3,15]. Hoff and Bashir synthesized empirical research into a three-layer model consisting of dispositional, situational, and learned trust factors [15]. This is particularly useful for autonomous research because a scientist's initial trust may derive from general attitudes toward AI, whereas later trust is shaped by direct experience with a particular research agent.

2.2 Trust Calibration and Appropriate Reliance

Trust calibration is closely connected to reliance but should not be equated with it. A person may rely on an AI system because it is convenient, because of time pressure, or because the user lacks sufficient expertise to evaluate its output. Such reliance does not necessarily indicate calibrated trust.

Brzowski and Nathan-Roberts' systematic review of trust measurement in human-automation interaction found substantial methodological heterogeneity. Among the reviewed studies, subjective trust measures were common, while validated scales were used inconsistently [16]. This observation has become increasingly important in human-AI research, where researchers sometimes use self-reported trust as a proxy for successful human-AI collaboration.

Recent work explicitly argues for moving from trust measurement toward **appropriate reliance measurement**. Raees and Papangelis distinguish trust from objective reliance behavior and note that the literature remains fragmented regarding how appropriate reliance should be operationalized [17]. For autonomous research, this distinction is critical. A scientist reporting high trust in an AI research agent does not necessarily demonstrate appropriate reliance. The stronger outcome is whether the scientist accepts correct recommendations and appropriately challenges incorrect ones.

2.3 Human-AI Teaming and Mental Models

Human-AI collaboration differs from conventional automation because AI systems increasingly perform tasks associated with reasoning and knowledge work. Bansal et al. demonstrated that team performance depends not only on model accuracy but also on the human's mental model of the AI's error boundary—namely, understanding when the AI is likely to succeed or fail [18]. This insight is directly applicable to research agents.

A researcher does not need to know every internal detail of an AI model. Rather, the researcher needs an operational mental model such as:

- the agent is strong at searching and organizing large bodies of literature;
- it may miss papers outside its retrieval scope;
- it can synthesize evidence but may incorrectly infer causal relationships;
- it can generate code but does not guarantee statistical validity;
- it may express confidence without possessing reliable epistemic uncertainty;
- it should not be treated as an independent scientific authority.

Shared mental models are also important. Theoretical work on human-AI teams suggests that successful teaming requires humans and AI systems to maintain compatible representations of tasks, roles, capabilities, and coordination requirements [19]. In autonomous research, this means the researcher and agent should have a sufficiently aligned understanding of research objectives, evidence standards, stopping criteria, and responsibility boundaries.

2.4 Transparency and Explainability

Transparency is frequently proposed as a means of supporting trust calibration. Agent transparency includes information about what an agent is doing, why it is doing it, what information it is using, and what it intends to do next [20]. However, transparency is not automatically beneficial.

Excessive information can increase cognitive workload and obscure the most important signals. Conversely, insufficient transparency can make it impossible for researchers to detect errors or understand why an agent reached a conclusion. Therefore, transparency should be **decision-relevant** rather than simply comprehensive.

For scientific agents, useful transparency may include:

1. source provenance;
2. evidence supporting a claim;
3. uncertainty estimates;
4. assumptions;

5. tool calls;
6. failed searches or experiments;
7. alternative hypotheses;
8. code and computational environments;
9. conflicts between sources;
10. reasons for escalating a decision to a human.

The objective is not to expose every internal model state but to provide information that improves the human's ability to make appropriate reliance decisions.

3. Autonomous Research as a Human-Agent Collaboration Problem

3.1 From AI Assistance to Agentic Research

LLM-based autonomous agents generally combine a foundation model with memory, planning, tools, environmental interaction, and feedback mechanisms [8]. In scientific research, these components can be organized around a research workflow:

Research Goal→Literature→Hypothesis→Experiment→Analysis→Evaluation→Revision\text{Research Goal} \rightarrow \text{Literature} \rightarrow \text{Hypothesis} \rightarrow \text{Experiment} \rightarrow \text{Analysis} \rightarrow \text{Evaluation} \rightarrow \text{Revision}

An agent may operate at one or multiple stages. Agentic-science surveys characterize emerging systems according to their capabilities for literature analysis, hypothesis generation, experimentation, evaluation, and iterative discovery [9,10].

The resulting human-agent relationship is different from traditional decision support. A decision-support system might recommend an action while the human performs it. An autonomous research agent can potentially select subtasks, call tools, write files, execute experiments, interpret outputs, and decide what to investigate next.

Consequently, trust calibration must operate at multiple levels:

- **output level:** Can I trust this specific answer?
- **task level:** Can I trust the agent to perform this research task?
- **workflow level:** Can I trust the agent to complete this sequence?
- **system level:** Can I trust the agent's overall behavior?
- **delegation level:** How much authority should I give it?

3.2 Error Propagation in Autonomous Research

Autonomous research introduces a major challenge: errors can propagate across agentic workflows.

For example:

Incorrect Literature Interpretation→Incorrect Hypothesis→Invalid Experiment→Misleading Result→Incorrect Scientific Claim
 $\text{Incorrect Literature Interpretation} \rightarrow \text{Incorrect Hypothesis} \rightarrow \text{Invalid Experiment} \rightarrow \text{Misleading Result} \rightarrow \text{Incorrect Scientific Claim}$

The probability of a successful workflow therefore cannot be assumed to equal the accuracy of each individual component in isolation. Even if each stage has relatively high reliability, sequential dependencies can reduce end-to-end reliability.

This problem is intensified when an agent treats its own earlier outputs as evidence for subsequent reasoning. A false assumption generated during literature synthesis may become embedded in a research plan and subsequently appear as an established fact.

LLM hallucinations are particularly important here. Reviews have identified hallucination as a persistent problem in language-model generation, while empirical studies have demonstrated that models can produce unsupported or fabricated scholarly references [12,13]. Consequently, trust calibration in autonomous research must account for **uncertainty propagation**, not merely individual answer accuracy.

3.3 Trust and the Scientific Division of Labor

The most productive human-agent relationship is unlikely to involve either complete human control or unrestricted machine autonomy. Instead, the system should allocate responsibilities according to comparative strengths.

Humans remain particularly important for:

- defining research significance;
- identifying ethical constraints;
- interpreting ambiguous findings;
- assessing societal implications;
- deciding whether evidence is sufficient for consequential claims;
- recognizing domain-specific anomalies;
- establishing publication and authorship responsibilities.

AI agents may provide advantages in:

- large-scale literature screening;
- information organization;
- repetitive coding;
- simulation;
- hypothesis-space exploration;
- cross-domain association;
- rapid generation and comparison of alternatives.

This suggests that trust calibration should be connected to **capability-aware task allocation**. The relevant question is not simply "Do I trust the AI?" but "Which parts of this research process should I delegate to this agent under these conditions?"

4. Main Technical and Thematic Discussion

4.1 A Conceptual Model of Trust Calibration for Autonomous Research

A useful framework for autonomous research can represent calibrated collaboration as a function of six interacting variables:

$$TC=f(C,U,R,T,H,E) \quad TC = f(C, U, R, T, H, E)$$

where:

- CC = demonstrated agent capability;
- UU = uncertainty;
- RR = task and consequence risk;
- TT = transparency and provenance;
- HH = human expertise and workload;
- EE = interaction history and evidence.

The purpose is not to calculate a universal numerical trust score but to define the variables that should influence delegation and oversight.

Capability concerns how reliably the agent performs the particular task. An agent may be highly capable at summarization but poorly calibrated for causal inference.

Uncertainty concerns both epistemic uncertainty and uncertainty caused by incomplete information, retrieval limitations, ambiguity, or stochastic model behavior.

Risk refers to the consequences of error. A wrong formatting decision is low risk; an incorrect biomedical recommendation can be extremely high risk.

Transparency determines how easily a human can verify the agent's work.

Human expertise affects the ability to detect errors. Experts may challenge an AI's incorrect hypothesis more effectively than novices.

Interaction history captures evidence accumulated over time. A researcher may increase delegation after repeated successful performance or reduce it following observed failures.

This model implies that trust should be dynamic rather than static.

4.2 Over-Trust and Under-Trust

Two failure modes dominate trust calibration.

Over-trust

Over-trust occurs when the human relies on the agent beyond its demonstrated capability. In research, possible consequences include:

- accepting fabricated references;
- assuming generated hypotheses are novel;
- accepting incorrect statistical analyses;
- overlooking inappropriate experimental assumptions;
- treating fluent explanations as evidence of correctness.

LLMs can be especially susceptible to this problem because linguistic fluency can create an impression of competence. Research on hallucination demonstrates that models may produce confident but unsupported statements [12,13]. Thus, linguistic confidence should not be treated as epistemic confidence.

Under-trust

Under-trust occurs when researchers reject useful AI assistance despite adequate evidence of capability. This can reduce productivity and increase workload. Excessive skepticism can be particularly problematic when the agent is objectively better at a narrow task, such as searching a large literature corpus or exploring many computational configurations.

Appropriate calibration therefore requires neither skepticism nor enthusiasm as a universal principle. It requires **evidence-sensitive reliance**.

4.3 Capability and Error-Boundary Communication

Bansal et al. showed that human-AI team performance improves when users understand the AI's error boundary [18]. For autonomous research, interfaces should therefore communicate not merely aggregate accuracy but **where and when the agent is likely to fail**.

For example, an agent might report:

High reliability: bibliographic metadata extraction from verified sources.

Moderate reliability: synthesis across heterogeneous studies.

Low reliability: determining whether a hypothesis is genuinely novel without comprehensive literature coverage.

Such capability profiles can be more useful than a generic "confidence: 92%" indicator.

A capability profile could be represented as:

$$P(A|D,T)P(A|D,T)$$

where AA represents agent success, DD represents the domain, and TT represents the task type. The system should ideally estimate capability conditional on the specific research context rather than presenting a single global reliability score.

4.4 Provenance as a Trust-Calibrating Mechanism

Scientific research requires traceability. An autonomous research agent should therefore preserve a provenance chain connecting conclusions to evidence.

A provenance graph can be represented as:

Claim→Evidence→Source→Method→Result
Claim \rightarrow Evidence \rightarrow Source
\rightarrow Method \rightarrow Result

For example, if an agent proposes a hypothesis, the researcher should be able to inspect:

1. which papers influenced the hypothesis;
2. which findings were treated as established;
3. which assumptions were introduced by the agent;
4. whether contradictory evidence was considered;
5. which computational or experimental results support the proposal.

Provenance can reduce the burden of verification because researchers can focus attention on high-risk reasoning steps rather than reconstructing the entire workflow.

4.5 Uncertainty Communication

Uncertainty is one of the most important mechanisms for trust calibration. However, probability values alone may be misleading if they are not empirically calibrated.

For autonomous research agents, uncertainty should be multidimensional. A useful distinction is:

$U = (U_{data}, U_{model}, U_{retrieval}, U_{reasoning}, U_{execution})$
 $U = (U_{\{data\}}, U_{\{model\}}, U_{\{retrieval\}}, U_{\{reasoning\}}, U_{\{execution\}})$

where:

- U_{data} : uncertainty due to data quality;
- U_{model} : uncertainty about model behavior;
- $U_{retrieval}$: uncertainty caused by incomplete or ambiguous evidence retrieval;
- $U_{reasoning}$: uncertainty in inference;
- $U_{execution}$: uncertainty about tools, code, or experimental execution.

This is preferable to a single confidence score because a conclusion can be highly certain given the retrieved data while the retrieval itself is incomplete.

Methods such as semantic entropy have demonstrated potential for detecting uncertainty associated with semantically different model outputs [14]. Such approaches may contribute to research-agent oversight, although they should not be treated as complete solutions to scientific uncertainty.

4.6 Adaptive Oversight

Fixed human approval checkpoints create a tension between safety and efficiency. Requiring researchers to approve every action can turn an autonomous agent into an expensive interface for manual supervision. Conversely, allowing unrestricted execution can create unacceptable risks.

A better approach is **adaptive oversight**.

One conceptual policy is:

$$I = \begin{cases} 0, & R \cdot U < \tau_1 \\ 1, & \tau_1 \leq R \cdot U < \tau_2 \\ 2, & R \cdot U \geq \tau_2 \end{cases}$$

where I represents the level of human intervention and τ_1, τ_2 are calibrated thresholds.

For example:

- **Level 0 — autonomous execution:** low-risk, high-confidence tasks;
- **Level 1 — asynchronous review:** moderate-risk tasks requiring later inspection;
- **Level 2 — mandatory approval:** high-risk or high-uncertainty tasks.

Such policies align oversight intensity with risk rather than treating all actions equally.

Recent work on meaningful human oversight similarly argues that human involvement should not become a superficial "rubber stamp" and that oversight architectures must preserve meaningful human agency [21].

5. Current Approaches and Developments

5.1 Human-in-the-Loop Research Agents

Human-in-the-loop systems retain human decision authority at predefined stages. This approach is comparatively straightforward and is appropriate when consequences are high or verification is feasible.

Its primary limitation is scalability. Autonomous research may involve hundreds or thousands of intermediate decisions, making complete human review impractical.

5.2 Human-on-the-Loop and Supervisory Models

Human-on-the-loop architectures allow agents to act autonomously while humans supervise the process. Humans intervene when predefined conditions are violated.

This architecture is better suited to long-running research workflows but places greater demands on monitoring interfaces and anomaly detection.

The main challenge is that passive monitoring can result in reduced vigilance. If an agent performs many successful actions, humans may become less attentive precisely when a rare but consequential failure occurs.

5.3 Multi-Agent Scientific Systems

Recent systems increasingly use multiple specialized agents. The AI co-scientist, for example, employs specialized components for generation, reflection, ranking, evolution, proximity analysis, and meta-review [7]. Multi-agent designs can provide diversity of reasoning and internal critique.

However, multi-agent systems introduce a new trust problem: **Should the human trust the system, each agent, or the aggregation mechanism?**

Research on multi-agent trust suggests that users may generalize trust across components rather than evaluating each agent independently. Walliser, de Visser, and Shaw found that a malfunctioning agent can negatively affect trust in other agents, while transparency and training influence system-wide trust [22].

Therefore, a multi-agent research platform should expose agent roles, provenance, disagreement, and evidence rather than presenting a single undifferentiated "AI researcher."

5.4 AI-Generated Critique and Self-Verification

Agents can use one model or agent to generate an answer and another to critique it. Debate, reflection, self-consistency, and iterative evaluation can reduce some classes of errors.

The AI co-scientist exemplifies this approach through generation, debate, ranking, and evolutionary refinement [7]. The AI Scientist similarly uses automated evaluation as part of its research workflow [6].

Nevertheless, self-verification does not eliminate the need for human oversight. If multiple agents share the same underlying biases, training data, or reasoning weaknesses, agreement among them may create an illusion of independent validation.

5.5 Retrieval-Augmented and Evidence-Grounded Agents

Retrieval-augmented systems can ground agent responses in external documents. In scientific research, this is valuable because claims can be connected to papers and datasets.

However, retrieval grounding does not automatically guarantee correctness. A system can retrieve an inappropriate source, misinterpret a correct source, overlook contradictory evidence, or draw conclusions not justified by the cited material.

Therefore:

Retrieved Evidence $\neq$ Validated Evidence $\text{\texttt{\text{Retrieved Evidence}}} \neq \text{\texttt{\text{Validated Evidence}}}$

Evidence grounding should be complemented by source verification, contradiction detection, provenance tracking, and human review for consequential claims.

5.6 Current Autonomous Scientific Discovery Systems

The AI Scientist represents one of the clearest demonstrations of end-to-end research automation. It can generate research ideas, write and execute code, produce visualizations, and generate scientific papers [6]. Its significance is not that it eliminates scientists, but that it demonstrates how much of the research workflow can be computationally orchestrated.

The AI co-scientist takes a more explicitly collaborative approach. It is designed to work with scientists, using specialized agents to generate and refine hypotheses while incorporating scientist-provided goals and feedback [7].

Recent surveys describe this broader development as a shift toward "agentic science," in which AI systems increasingly participate in multiple stages of the scientific discovery lifecycle [9,10]. These developments make trust calibration a foundational design problem rather than a secondary usability concern.

6. Research Challenges and Limitations

6.1 Lack of Standardized Trust-Calibrated Evaluation

Current studies use heterogeneous definitions and measurements of trust and reliance [16,17]. There is no widely accepted benchmark for asking whether a scientific agent has produced **appropriately calibrated human reliance**.

Future evaluations should separately measure:

- trust;
- reliance;
- appropriate reliance;
- error detection;
- intervention quality;
- delegation decisions;
- scientific task performance.

6.2 Difficulty Measuring Scientific Correctness

Scientific correctness is often delayed. A hypothesis may appear plausible but require months of experimental validation. This makes real-time trust calibration difficult.

A research agent could therefore be evaluated on multiple layers:

Performance = $f(\text{factuality}, \text{methodological validity}, \text{reproducibility}, \text{novelty}, \text{scientific utility})$

No single metric is sufficient.

6.3 Compound and Cascading Errors

Autonomous workflows can transform small errors into large downstream failures. Current trust models often evaluate individual interactions rather than long-horizon workflows.

Research must therefore investigate **trajectory-level reliability** rather than isolated response accuracy.

6.4 Human Cognitive Workload

More transparency does not necessarily mean better collaboration. If researchers must inspect every citation, tool call, intermediate hypothesis, and reasoning step, cognitive workload can become excessive.

Trust-calibration interfaces should therefore prioritize information according to expected decision value.

6.5 Automation Bias and Anchoring

Humans can anchor on AI recommendations even when independent judgment is possible. Once an agent proposes a hypothesis, researchers may evaluate alternatives relative to that proposal rather than generating independent hypotheses.

This creates a particularly important issue for scientific creativity: an AI agent may increase research efficiency while unintentionally narrowing the hypothesis space.

6.6 Expertise Differences

Experts and novices may calibrate differently. Experts possess domain knowledge that allows them to identify implausible claims, whereas novices may depend heavily on the agent.

Thus, the same interface may produce appropriate reliance for experts but over-reliance among inexperienced researchers.

6.7 Multi-Agent Trust

As systems become collections of specialized agents, the human must understand not only whether an output is reliable but also which component produced it and whether the components independently support one another.

This problem remains underexplored compared with single-agent trust [22].

6.8 Longitudinal Trust Dynamics

Trust evolves through experience. A researcher may initially distrust an agent and subsequently develop excessive confidence after many successful interactions.

Most experiments are short-term, making it difficult to understand how trust evolves over months of scientific collaboration.

7. Research Gaps and Future Directions

7.1 Trust Calibration Benchmarks for Scientific Agents

A major research priority is the development of standardized benchmark environments in which researchers collaborate with autonomous agents on realistic scientific tasks.

Such benchmarks should contain:

- correct and incorrect literature claims;
- fabricated citations;
- incomplete evidence;
- conflicting papers;
- statistical errors;
- flawed code;
- plausible but invalid hypotheses;
- varying task risk.

Researchers could then be evaluated on whether they accept correct agent outputs and reject incorrect ones.

7.2 From Trust Scores to Reliance Policies

Future systems should avoid treating "trust" as a single scalar. Instead, they should estimate whether the human's **current delegation policy** is appropriate.

A more useful research objective is:

$$\text{Optimal Reliance} = \arg\max_d [U(d) - \lambda \text{Risk}(d)]$$

$\text{Optimal Reliance} = \arg\max_d [U(d) - \lambda \text{Risk}(d)]$

where d is the delegation decision, $U(d)$ is expected scientific utility, and $\text{Risk}(d)$ represents the consequences of inappropriate delegation.

This reframes trust calibration as decision optimization.

7.3 Adaptive Trust Calibration

Agents should learn from human corrections without simply learning to become more agreeable.

For example, if a researcher repeatedly corrects an agent's interpretation of a specific experimental domain, the system should update its capability model and lower its autonomous authority for related tasks.

The agent could maintain a task-specific reliability profile:

$$\text{Reliability}_{t+1} = \text{Reliability}_t + \alpha (\text{ObservedSuccess} - \text{ExpectedSuccess})$$

$\text{Reliability}_{t+1} = \text{Reliability}_t + \alpha (\text{ObservedSuccess} - \text{ExpectedSuccess})$

This resembles adaptive learning but should be constrained to prevent temporary failures from causing disproportionate reductions in autonomy.

7.4 Shared Mental Models

Future research should investigate whether autonomous research agents can explicitly maintain models of:

- researcher expertise;
- research objectives;
- acceptable evidence standards;
- risk tolerance;
- preferred intervention thresholds;
- known agent limitations.

Conversely, interfaces should help researchers construct accurate mental models of the agent.

The objective is a **shared human-agent model of the research workflow**, extending existing shared-mental-model theory to autonomous scientific systems [19].

7.5 Provenance-Centered Scientific Interfaces

Future research interfaces should make provenance a first-class interaction mechanism. Rather than merely showing an answer, the interface should expose the evidence chain behind important conclusions.

Researchers could interact directly with the provenance graph:

Claim → **supporting sources** → **contradictory sources** → **reasoning step** → **experiment** → **result**

Such interfaces could enable selective verification and reduce unnecessary cognitive workload.

7.6 Calibration of Agent Confidence

Agent confidence should be empirically tested against actual outcomes. A system that expresses high confidence for correct answers and low confidence for incorrect ones is more useful for human oversight than one that is uniformly confident.

Calibration metrics such as expected calibration error may be adapted, but research agents require richer evaluation because uncertainty exists at multiple workflow stages.

7.7 Human Control as a Dynamic Resource

Human oversight should be treated as a limited cognitive resource. The system should allocate human attention to decisions where it has the highest expected value.

For example:

- routine literature deduplication → autonomous;
- summarization of established findings → low oversight;
- interpretation of conflicting studies → moderate oversight;
- selection of a high-impact experiment → high oversight;
- publication of a consequential scientific claim → mandatory expert verification.

This model preserves both efficiency and meaningful human agency.

7.8 Independent Verification and Diversity

A critical future direction is developing mechanisms for genuine independent verification. Multiple agents that share the same model family cannot necessarily be considered independent evaluators.

Research should explore heterogeneous verification involving:

- different foundation models;
- symbolic tools;
- statistical validators;
- domain-specific simulators;
- external databases;
- human experts.

Agreement across genuinely independent evidence sources should carry more weight than agreement among several instances of the same model.

7.9 Ethical and Governance Implications

Autonomous research raises questions concerning responsibility and authorship. If an agent generates a hypothesis, writes code, conducts an analysis, and drafts a manuscript, the human researcher's responsibility cannot be reduced simply because the agent performed the technical work.

Human oversight should therefore include explicit accountability mechanisms. Recent work on meaningful human oversight emphasizes that human involvement must retain substantive decision authority rather than becoming a formal approval step [21].

8. Proposed Research Agenda

Based on the literature reviewed above, a practical research agenda for human trust calibration in autonomous research can be organized into five layers.

Layer 1: Agent Reliability

Measure the agent's task-specific performance under controlled conditions.

Layer 2: Human Mental Model

Measure whether researchers understand the agent's strengths, weaknesses, and error boundaries.

Layer 3: Reliance Behavior

Measure whether researchers accept correct outputs and override incorrect outputs.

Layer 4: Workflow-Level Collaboration

Evaluate performance across complete research trajectories rather than isolated interactions.

Layer 5: Long-Term Scientific Outcomes

Measure effects on research productivity, reproducibility, novelty, quality, and human cognitive workload.

This layered approach avoids a common methodological problem in which high self-reported trust is interpreted as evidence of successful collaboration.

A future benchmark could therefore report a vector:

$$B=(A,C,R,M,W,S) \quad B = (A, C, R, M, W, S)$$

where:

- AA = agent accuracy/reliability;
- CC = calibration;
- RR = appropriate reliance;
- MM = mental-model accuracy;
- WW = human workload;
- SS = scientific outcome quality.

Such a benchmark would provide a substantially richer evaluation than a single trust score.

9. Conclusion

Human trust calibration is likely to become one of the central human-factors problems in autonomous scientific research. As AI systems move from assisting individual research tasks toward coordinating multi-step research workflows, the consequences of inappropriate reliance become increasingly significant. Traditional automation research establishes that the objective should not be maximum trust but **appropriate reliance**, determined by the relationship between human expectations and system capability [1,2]. Human-AI teaming research further demonstrates that accurate mental models of AI capabilities and error boundaries are essential for effective collaboration [18,19].

Autonomous research intensifies these challenges because scientific workflows are open-ended, evidence is incomplete, and errors can propagate across multiple stages. Current

systems such as The AI Scientist and AI co-scientist demonstrate the growing technical feasibility of autonomous or semi-autonomous scientific workflows [6,7]. At the same time, hallucinations, uncertain reasoning, incomplete retrieval, and workflow-level error propagation demonstrate why human verification remains necessary [12–14].

The future of autonomous research should therefore not be framed as a choice between human control and machine autonomy. A more productive paradigm is **calibrated delegation**: humans delegate tasks according to demonstrated agent capability, task risk, uncertainty, and the availability of verification. Agents should communicate limitations and provenance, adapt their level of autonomy based on evidence, and escalate consequential decisions to humans.

The most important research gap is consequently not simply how to make AI agents more accurate. It is how to create **human-agent systems in which both capability and reliance are calibrated over time**. Progress will require standardized appropriate-reliance benchmarks, longitudinal human-subject studies, adaptive oversight mechanisms, provenance-centered interfaces, uncertainty-aware agents, and shared mental models.

Ultimately, trustworthy autonomous research should not produce humans who trust AI blindly. It should produce human researchers who know **when to trust, when to question, when to verify, and when to take control**.

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Note on source quality. The paper prioritizes peer-reviewed journal and conference literature for foundational claims and clearly identifies recent arXiv/preprint work where the rapidly developing autonomous-science literature has not yet accumulated equivalent peer-reviewed coverage. Bibliographic details and persistent identifiers were checked against scholarly or publisher/index records where available. The 2026 literature is included primarily to characterize the current research frontier rather than to substitute for established foundational work.